

Started on Friday, 10 January 2025, 11:56 AM**State** Finished**Completed on** Friday, 10 January 2025, 1:20 PM**Time taken** 1 hour 24 mins**Grade** Not yet graded**Question 1**

Correct

Mark 2.00 out of 2.00

Select, for the following techniques, if this is meant to improve the final model generalization capabilities or the performance of training procedure (backpropagation). With performance of backpropagation we mean also convergence speed and training stability.

generalization	performance		
☐	☒	Batch normalization	✓
☒	☐	Data augmentation	✓
☒	☐	Early stopping	✓
☒	☐	Weight decay	✓
☐	☒	Gradiente descent with momentum	✓
☐	☒	Xavier weight initialization	✓

Batch normalization: performance

Data augmentation: generalization

Early stopping: generalization

Weight decay: generalization

Gradiente descent with momentum: performance

Xavier weight initialization: performance

Question 2

Partially correct

Mark 0.50 out of 1.00



Bruno Zamborlin demonstrates a HyperSurfaces-enabled car door, where taps, touches and knocks are registered by vibration sensors and transformed into actions by AI algorithms running on a standalone system-on-chip.

Select which of the following is correct for a possible implementation of this technology.

- ☐ a. A recurrent neural network taking the sequence of recordings from the vibration sensors as input. Using a softmax output layer we can select the action to be performed. However, by this model, **we can count the knocks, but we cannot discriminate the position of the touch or the knock.**
- ☐ b. A 1D convolutional neural network. The input is a 1D signal with multiple channels; each channel is the signal from one vibration sensor. The 1D convolution is applied in a windowed manner, i.e., on a chunk containing few seconds of data. The model is trained via categorical cross-entropy. This model **can both count the knocks and discriminate the position of the touch or the knock.**
- ☒ c. A feedforward neural network for classification. Each class corresponds to one out of a list of possible actions. The input of the network is the vibration intensity captured by the vibration sensors at each time instant. However, by this model, **we can only discriminate the position of the touch or the knock, but not count the knocks.** ✓

Your answer is partially correct.

You have correctly selected 1.

The correct answers are:

A feedforward neural network for classification. Each class corresponds to one out of a list of possible actions. The input of the network is the vibration intensity captured by the vibration sensors at each time instant. However, by this model, **we can only discriminate the position of the touch or the knock, but not count the knocks.**,

A 1D convolutional neural network. The input is a 1D signal with multiple channels; each channel is the signal from one vibration sensor. The 1D convolution is applied in a windowed manner, i.e., on a chunk containing few seconds of data. The model is trained via categorical cross-entropy. This model **can both count the knocks and discriminate the position of the touch or the knock.**

Question 3

Complete

Marked out of 3.00

Attention (used in Seq2seq models) and Self-Attention (used in Transformers) are two powerful mechanisms to improve the performance of models for sequential data. Provide a short answer to the following questions making sure you keep them separate in the answer (as we grade them independently 1 point each).

a) What is the difference between attention and self attention?

b) What is the difference between dot product, Luong, and Bahdanau scoring mechanisms in terms of final model parameters?

c) What is the encoder-decoder attention? How it works?

Keep the answers short as answering things which are not related to the question get penalized.

a) attention is used in LSTM type of models to give certain importance to different parts of the input. attention mechanism still needs the memory cell to have the short term info. Self-attention is used as a layer in transformers where attention scores are calculated based on the transformation matrices of the input embeddings Q,K,V that builds the long term dependencies over the input to build the context vector.

b) dot product is the simplest one compared to the other two, it has lower number of parameters and easier to be computed.

c) in transformers, in the decoder part besides having self-attention layer and FFNN (that are present in the encoder part), there is also encoder-decoder attention. This ensures that the decoder grabs attention also before the complete embedding of the input (from different parts of the encoder) to make its output. This is done with different connections throughout the network exploiting the multi-head attention structure.

Question 4

Correct

Mark 2.00 out of 2.00

We are in the general setting of regression, which of the following statements holds true (independently from the others).

- ☒ a. I regularize the model by adding the sum of absolute values of the weights (L1 regularization) to the loss function. ✔ Yes this is a typical mean of regularization
- ☐ b. I use the categorical cross-entropy loss function
- ☒ c. If I normalize the output to be in the $[-1, +1]$ range I can use a hyperbolic tangent activation function ✔ Yes, it might have some saturation problems but provided you use a regression loss it is perfectly fine
- ☒ d. I use the mean absolute error loss function ✔ Yes it is one of the possible losses for a Regression task
- ☐ e. I use the mean absolute error as loss function to force only positive output values
- ☐ f. In a multi valued regression task I use as many output as the number of values to regress and the softmax activation function
- ☒ g. In a multi valued regression task I can use as many output as the number of values to regress and the ReLu activation function to force only positive outputs ✔ Yes, linear activations are used in regression, I can use ReLu if I know that the domain of the regression is positive

Your answer is correct.

The correct answers are:

I use the mean absolute error loss function,

In a multi valued regression task I can use as many output as the number of values to regress and the ReLu activation function to force only positive outputs,

I regularize the model by adding the sum of absolute values of the weights (L1 regularization) to the loss function.,

If I normalize the output to be in the $[-1, +1]$ range I can use a hyperbolic tangent activation function

Question 5

Correct

Mark 2.00 out of 2.00

Select among the following which statements are true. The statements should be considered independently.

True	False		
☒	☐	Word embedding is used to transform one hot encoding word representation into a dense vector representation.	✓
☒	☐	Word embedding training can be considered as an unsupervised learning task.	✓
☐	☒	In the word2vec model the embedding space regularity, i.e., constant difference vector for similar concepts, is forced explicitly at training time with a specific loss.	✓
☒	☐	Skip grams and Continuous Bag of Words are two possible implementations of the word2vec word embedding model.	✓

Word embedding is used to transform one hot encoding word representation into a dense vector representation.: True

Word embedding training can be considered as an unsupervised learning task.: True

In the word2vec model the embedding space regularity, i.e., constant difference vector for similar concepts, is forced explicitly at training time with a specific loss.: False

Skip grams and Continuous Bag of Words are two possible implementations of the word2vec word embedding model. : True

Question 6

Partially correct

Mark 0.25 out of 2.00

Consider a MLP to classify images over 10 classes. The input layer of the network is directly connected to a flattening and then to the output layer (no hidden layer).

Mark all the following answers that are true. Wrong answers result in negative points.

Select one or more:

- ☒ a. This is a classifier for hand-crafted features. ✗ FALSE: there is no feature extraction
- ☐ b. The latent representation computed by this network can be used to perform image retrieval.
- ☒ c. The network is equivalent to a 1D CNN having a single convolutional layer, which classifies images as time series. ✗ FALSE: this is not a CNN, it has a dense connectivity
- ☐ d. Since the network is not convolutional, there is no point of using translation as a transformation for augmentation during training.
- ☒ e. Once the network is trained, the input image is assigned to the class corresponding to the closer template in terms of Euclidean distance. ✗ FALSE: correlation is computed instead
- ☐ f. During training, the network learns a set of images having the same dimension as the input image, and then computes the correlation between these images and the input to define the output.
- ☐ g. Since the image is unfolded in a vector, the same network can classify images with a number of channels that is different from the training images.
- ☒ h. The network needs to have a softmax or a sigmoid activation in the output neurons, if we train the network using cross entropy loss ✓ TRUE: without any of these activations we won't have a probability estimate to compute these loss functions.
- ☐ i. Right before the nonlinearity in the output neurons, the network computes a set of scores associated with each input image. However, the predicted class might not correspond to the class yielding the largest score.
- ☐ j. This network is equivalent to using two linear classifiers followed by nonlinearities, in cascade.

Your answer is partially correct.

You have selected too many options.

The correct answers are:

During training, the network learns a set of images having the same dimension as the input image, and then computes the correlation between these images and the input to define the output.

,

The network needs to have a softmax or a sigmoid activation in the output neurons, if we train the network using cross entropy loss functions.

Question 7

Incorrect

Mark 0.00 out of 1.00

What is the receptive field of this network? Consider the network takes as input an image 500x500

```
def receptive_field_2(input_shape):  
  
    input_layer = tfkl.Input(shape=input_shape, name='input_layer')  
  
    x = tfkl.Conv2D(16, kernel_size=3, padding='valid', activation='relu')(input_layer)  
    x = tfkl.MaxPooling2D(4)(x)  
    x = tfkl.Conv2D(32, kernel_size=5, padding='valid', activation='relu')(x)  
    x = tfkl.MaxPooling2D(3)(x)  
    x = tfkl.Conv2D(64, kernel_size=7, padding='valid', activation='relu')(x)  
    x = tfkl.MaxPooling2D(2)(x)  
    x = tfkl.Conv2D(128, kernel_size=9, padding='valid', activation='relu')(x)  
    x = tfkl.MaxPooling2D(1)(x)  
  
    model = tf.keras.Model(inputs=input_layer, outputs=x)  
  
    return model
```

Select one:

- ☐ a. 330
- ☐ b. The receptive field equals the entire input image, whatever the input is
- ☐ c. 106
- ☐ d. 306
- ☒ e. None of the others ✖ FALSE:
- ☐ f. 114
- ☐ g. 46
- ☐ h. It cannot be computed because the convolutions have valid padding

Your answer is incorrect.

The correct answer is:

306

Question 8

Partially correct

Mark 1.20 out of 2.00

Consider a semantic segmentation network suitably trained and mark all the answers that are true.

Wrong answers result in negative points.

Select one or more:

- ☐ a. In a semantic segmentation network, the last layer needs to be a conv2D layer with filters having size 1x1
- ☐ b. The U-Net is among the most popular segmentation network, but it can be used on medical images only.
- ☒ c. In a U-Net, the number of filters in the last layer needs to be equal to the number of classes. ✓ TRUE
- ☒ d. Semantic segmentation network can be trained using class-specific weights. ✓ TRUE: they are assigned to pixel values.
- ☐ e. When training a U-Net architecture, it is not possible to use photometric augmentation functions, otherwise we might lose correspondence with the label on the non-augmented image.
- ☐ f. The U-Net steadily increases the number of filters in each layer, to increase the spatial resolution of activations.
- ☐ g. When training a semantic segmentation network, we can use random masking to reproduce the stochastic effect of mini-batch selection.
- ☐ h. Semantic segmentation networks can be used to count persons in an image.
- ☐ i. A U-net can be obtained by moving a classifier trained on image patches (using patch-wise labels) to a fully convolutional networks.
- ☒ j. Several semantic segmentation networks have a contractive and an expanding paths. ✓ TRUE
- ☐ k. It is not possible to perform transfer learning in a semantic segmentation network.
- ☐ l. The U-Net architecture can be tested on images that are larger than those in the training set.

Your answer is partially correct.

You have correctly selected 3.

The correct answers are:

In a U-Net, the number of filters in the last layer needs to be equal to the number of classes.,

Semantic segmentation network can be trained using class-specific weights.,

When training a semantic segmentation network, we can use random masking to reproduce the stochastic effect of mini-batch selection.,

Several semantic segmentation networks have a contractive and an expanding paths.,

The U-Net architecture can be tested on images that are larger than those in the training set.

Question 9

Not answered

Not graded

Consider the following code

```
import tensorflow as tf
tfkl = tf.keras.layers
tfk = tf.keras

def get_model_2(input_shape, output_number):
    # Define the input layer
    input_layer = tfkl.Input(shape=input_shape, name='input_layer')

    c1 = tfkl.Conv2D(32, kernel_size=5, padding='valid', activation='relu', name='c1')(input_layer)
    c1 = tfkl.BatchNormalization(name='bn1')(c1)
    c1 = tfkl.AveragePooling2D(pool_size = (4,4), name='ap')(c1)
    c1 = tfkl.Dropout(0.25, name='dp1')(c1)

    s1 = tfkl.Conv2D(64, kernel_size=1, padding='same', activation='relu', name='c2')(c1)
    s1 = tfkl.Conv2D(32, kernel_size=5, padding='same', activation='relu', name='c3')(s1)
    s1 = tfkl.Add(name='ad1')([c1, s1])

    c2 = tfkl.Conv2D(64, kernel_size=5, padding='same', activation='relu', name='c4')(s1)
    c2 = tfkl.BatchNormalization(name='bn2')(c2)
    c2 = tfkl.AveragePooling2D(pool_size = (4,4), name='ap2')(c2)
    c2 = tfkl.Dropout(0.25, name='dp2')(c2)

    s2 = tfkl.Conv2D(128, kernel_size=1, padding='same', activation='relu', name='c5')(c2)
    s2 = tfkl.Conv2D(64, kernel_size=5, padding='same', activation='relu', name='c6')(s2)
    s2 = tfkl.Add(name='ad3')([c2, s2])

    gmp = tfkl.GlobalMaxPooling2D(name='gmp')(s2)

    fc1 = tfkl.Dense(128, activation='relu', name='dense_1')(gmp)
    fc1 = tfkl.BatchNormalization(name='bn')(fc1)
    fc2 = tfkl.Dense(64, activation='relu', name='dense_2')(fc1)

    output_layer = tfkl.Dense(output_number, activation="linear", name='output_layer')(fc2)

    model = tf.keras.Model(inputs=input_layer, outputs=output_layer)

    model.compile(
        loss=tf.keras.losses.MeanAbsoluteError(),
        optimizer=tf.keras.optimizers.Adam(learning_rate=0.001),
        metrics=[
            tf.keras.metrics.MeanSquaredError(),
            tf.keras.metrics.RootMeanSquaredError()
        ]
    )

    return model

input_shape = (228,116,3)
output_number = 6
model = get_model_2(input_shape=input_shape, output_number=output_number)
model.summary()
```

The last command returns the following summary, which we provide for reference.
 Feel free to copy and paste this command in the textbox below to prepare your answer.
 Consider that this question won't be corrected, numbers need to be reported in the next one

Layer (type)	Output Shape	Param #	Connected to
=====			

```
input_layer (InputLayer)
c1 (Conv2D)
bn1 (BatchNormalization)
ap (AveragePooling2D)
dp1 (Dropout)
c2 (Conv2D)
c3 (Conv2D)
ad1 (Add)
c4 (Conv2D)
bn2 (BatchNormalization)
ap2 (AveragePooling2D)
dp2 (Dropout)
c5 (Conv2D)
c6 (Conv2D)
ad3 (Add)
gmp (GlobalMaxPooling2D)
dense_1 (Dense)
bn (BatchNormalization)
dense_2 (Dense)
output_layer (Dense)
```

Total params:

Trainable params:

Non-trainable params:

Question 10

Correct

Mark 3.00 out of 3.00

Remember that this question will be automatically rated, therefore you have to consider the following details very carefully:

- when the activations become a vector, these are stacked along the third dimension (the channel, C).
Therefore, if your latent representation becomes a 3 element vector, enter as sizes H:1 W:1 C:3.
- Enter digits **only** and do not use "." or "," or "-" or "[". Say write "1234" and not "1,234" or "1.234". Write "0", not "-" or "[".
- Do not enter formulas, just resulting numbers
- Do not enter empty spaces... please double check
- You don't need to fill in connections, but please consider them.
- Wrong answers result in negative points

All the numbers have to be entered here, the text in the previous question is just meant for you to design the solution.

Layer (type)	Output Shape	Param #	Connected to (omitted, not to be entered)
input_layer (InputLayer)	H: 228 ✓, W: 116 ✓, C: 3 ✓	Params #: 0 ✓	
c1 (Conv2D)	H: 224 ✓, W: 112 ✓, C: 32 ✓	Params #: 2432 ✓	
bn1 (BatchNormalization)	H: 224 ✓, W: 112 ✓, C: 32 ✓	Params #: 128 ✓	
ap (AveragePooling2D)	H: 56 ✓, W: 28 ✓, C: 32 ✓	Params #: 0 ✓	
dp1 (Dropout)	H: 56 ✓, W: 28 ✓, C: 32 ✓	Params #: 0 ✓	
c2 (Conv2D)	H: 56 ✓, W: 28 ✓, C: 64 ✓	Params #: 2112 ✓	
c3 (Conv2D)	H: 56 ✓, W: 28 ✓, C: 32 ✓	Params #: 51232 ✓	
ad1 (Add)	H: 56 ✓, W: 28 ✓, C: 32 ✓	Params #: 0 ✓	
c4 (Conv2D)	H: 56 ✓, W: 28 ✓, C: 64 ✓	Params #: 51264 ✓	
bn2 (BatchNormalization)	H: 56 ✓, W: 28 ✓, C: 64 ✓	Params #: 256 ✓	
ap2 (AveragePooling2D)	H: 14 ✓, W: 7 ✓, C: 64 ✓	Params #: 0 ✓	
dp2 (Dropout)	H: 14 ✓, W: 7 ✓, C: 64 ✓	Params #: 0 ✓	
c5 (Conv2D)	H: 14 ✓, W: 7 ✓, C: 128 ✓	Params #: 8320 ✓	
c6 (Conv2D)	H: 14 ✓, W: 7 ✓, C: 64 ✓	Params #: 204864 ✓	
ad3 (Add)	H: 14 ✓, W: 7 ✓, C: 64 ✓	Params #: 0 ✓	
gmp (GlobalMaxPooling2D)	H: 1 ✓, W: 1 ✓, C: 64 ✓	Params #: 0 ✓	
dense_1 (Dense)	H: 1 ✓, W: 1 ✓, C: 128 ✓	Params #: 8320 ✓	
bn (BatchNormalization)	H: 1 ✓, W: 1 ✓, C: 128 ✓	Params #: 512 ✓	
dense_2 (Dense)	H: 1 ✓, W: 1 ✓, C: 64 ✓	Params #: 8256 ✓	
output_layer (Dense)	H: 1 ✓, W: 1 ✓, C: 6 ✓	Params #: 390 ✓	
=====			
Total params:	338086 ✓		
Trainable params:	337638 ✓		
Non-trainable params:	448 ✓		

Question 11

Partially correct

Mark 1.75 out of 2.00

Mark all the answers that are true. When considering a task, assume that the network is provided with all the labeling information needed during training.

Please, refrain from marking answers that are "technically possible in principle in Python", and stick to those tasks that a Neural Network expert (as you are expected to be) would use this network for.

Wrong answers result in negative points.

Select one or more:

- ☐ a. The add layer in the network of the previous question is meant to increase the number of channels, and eventually the size of the latent representation.
- ☐ b. Given an histological image of many nuclei, the network performs instance segmentation.
- ☐ c. Given an input image of a newspaper page, the network counts the number of characters, the number of words and of the articles in that page.
- ☐ d. Given a picture of a car registered in Italy, the network reads the licence plate.
- ☒ e. Given an aerial image as input, the network forecasts, in each pixel, both the wind speed and the temperature in three different time instants (8.00 AM, 16.00, 24.00). ✗ FALSE: that would be a network returning an image
- ☐ f. The network estimates the average color of the input image.
- ☐ g. This is an object detection network, but can localize only objects belonging to a single class.
- ☐ h. Given a picture depicting an animal, the network classifies whether it is a mammal, a reptile, an insect, a fish, a shell or none of the above
- ☒ i. Given a picture depicting a cat, the network locates the position of the center of the pupils and the center of the tail. ✓ TRUE: it's a 6-output regression
- ☒ j. Given the picture of a professor, the network estimates his age, the number of MSc and BSc courses in the first and in the second semester, and his salary. ✓ TRUE: this is a 6 output regression
- ☒ k. Given a picture of a vehicle, the network returns the three spatial dimensions (height, width and lenght) as well as the barycenter in the 3D space around the vehicle. ✓ TRUE: the network can in principle perform 3D localization from a single camera, as long as the bounding box is axis parallel (only 6 outputs)

Your answer is partially correct.

You have selected too many options.

The correct answers are:

Given a picture depicting a cat, the network locates the position of the center of the pupils and the center of the tail., Given the picture of a professor, the network estimates his age, the number of MSc and BSc courses in the first and in the second semester, and his salary.

, Given a picture of a vehicle, the network returns the three spatial dimensions (height, width and lenght) as well as the barycenter in the 3D space around the vehicle.